

Final Report

Probability and Statistics Applied to Information Technology

Regulations:

You should solve and submit this report to your eLearning account within two weeks, from the beginning of December 15th 2024 to the end of December 28th 2024. Late submissions will be not accepted. Additionally, students must comply with the following regulations:

- This is a group final report. Number of students in a group must not exceed 2.
- Any case of plagiarism will get 0.
- Assignments must be well-organized and legible. Format violations and lack of care may incur a penalty of 10% to 50% of the total score.
- English is required for high-quality 100% English classes.

Submission:

Students need to submit a document file is in both Word and Pdf formats (.doc/docx and .pdf), named by your Student IDs, eg. 52200000_52200001.docx, using our faculty's format. Limit the answer to a maximum of 15 pages, excluding title pages, references, and table of contents.

Requirement:

Students need to adhere to the following requirements:

- For each question, students must reason, explain, and present in detail the calculation steps, including the formula, alternative values , and final results.
- A conclusion is necessary for each answer.

Question 1: (1.0 point)

Consider the following game. The player tosses a die once only. The payoff is 1 dollar for each dot on the upturned face. Assuming this is a fair 6-sided die and you are the host. How much should you charge for this game? Why?

Question 2: (1.0 points)

Consider a year with $n=365$ days, where each day is equally likely to be someone's birthday. Now, suppose there are $k=13$ people at a party.

- a) What is the probability that at least two people in the party share the same birthday? (0.5 point)
- b) If you are also attending the party, what is the probability that at least one person has the same birthday as you? (0.5 point)

Question 3: (1.0 points)

There are two boxes. The red box contains four red balls and one blue ball, and the blue box contains five blue balls and three red balls. The following experiment is conducted one n times: a ball is randomly drawn from one of the two boxes, its color is recorded, and then the ball is returned to the original box. The first ball is drawn from the red box, but each subsequent draw is determined by the color of the ball just drawn: if it is red, the next draw will be from the red box; if it is blue, the next draw will be from the blue box. Calculate the probabilities for the following events:

- a) The first ball is red. (0.25 point)
- b) The second ball is red. (0.25 point)
- c) Analyze and find a general solution for the n^{th} ball. (0.5 point)

Question 4: (1.0 point)

There is a bag containing 40 blue balls and 60 red balls. I randomly select 10 balls without replacement. Let X represent the number of blue balls and Y represent the number of red balls selected. Determine the joint probability mass function (PMF) of X and Y .

Question 5: (1.0 points)

A marksman fires 25 bullets at a target, one at a time. The probability of hitting the target on a single shot is 0.6, and each shot is independent of the others. For each hit, he earns 1 point, and for each miss, he loses 1 point. Calculate the probability that his total score is between 4 and 6 points.

Question 6: (1.0 points)

Given a random variable $X \sim \text{Bin}(3, \theta)$, observe the random sample (1,5,4,3,6,8). Using the method of Maximum Likelihood Estimation (MLE) to estimate the parameter θ .

Question 7: (1.5 points)

A study is conducted to analyze weekly grocery expenditures among families in a neighborhood. A random sample of 100 families is taken, and their weekly grocery expenses are recorded. Here is a sample of 100 weekly grocery expenses:

Expenses (\$): 63.0, 81.0, 76.0, 87.0, 80.0, 90.0, 89.0, 89.0, 79.0, 82.0, 82.0, 92.0, 80.0, 79.0, 88.0, 94.0, 80.0, 86.0, 88.0, 76.0, 93.0, 78.0, 81.0, 89.0, 82.0, 89.0, 87.0, 75.0, 89.0, 94.0, 76.0, 86.0, 93.0, 82.0, 93.0, 92.0, 80.0, 87.0, 86.0, 86.0, 89.0, 76.0, 84.0, 76.0, 83.0, 78.0, 87.0, 81.0, 85.0, 74.0, 89.0, 77.0, 80.0, 74.0, 87.0, 77.0, 80.0, 92.0, 86.0, 74.0, 84.0, 85.0, 85.0, 87.0, 97.0, 84.0, 92.0, 86.0, 77.0, 93.0, 90.0, 73.0, 73.0, 88.0, 70.0, 88.0, 78.0, 85.0, 74.0, 83.0, 85.0, 91.0, 85.0, 78.0, 81.0, 89.0, 79.0, 83.0, 80.0, 81.0, 87.0, 76.0, 78.0, 78.0, 70.0, 78.0, 93.0, 78.0, 83.0, 90.0

- Construct a 95% confidence interval for the mean weekly grocery expenditure. (0.5 point)
- Explain the meaning of a 95% confidence coefficient. (0.25 point)
- Explain how confidence interval width changes with increasing confidence level at a fixed sample size. Give a specific example. (0.5 point)
- If the distribution of weekly expenditures is not normal, would the confidence intervals obtained in part (a) still be valid? Explain why or why not. (0.25 point)

Question 8: (1.0 points)

Twelve college students adopted a new study technique for one week and subsequently took a series of tests that measured their thinking time, fluency, flexibility, and originality of thought. The overall test scores of these students were compared to the average scores expected from students using their usual study methods. Below is a sample dataset of the test scores for the 12 students:

[44, 60, 81, 36, 67, 82, 94, 68, 48, 67, 59, 51]

- a) Test the hypothesis that the actual mean score of students using the new study technique is less than 80, with a significance level of 5%. (0.5 point)
- b) To be valid, what assumption is required for the hypothesis test in part (a)? (0.5 point)

Question 9: (1.5 points)

This question asks you to differentiate between Probability and Statistics, classify the course topics accordingly, and visualize the entire course structure through a comprehensive mind map.

- a) Explain the difference between Probability and Statistics with clear definitions and examples. (0.5 point)
- b) Identify the topics in this course that belong to Probability and those that belong to Statistics. Provide a classification and briefly justify your reasoning. (0.5 point)
- c) Create a mind map summarizing all the key knowledge areas in the course. The mind map should organize topics under Probability and Statistics and highlight their interconnections. (0.5 point)

This is the end of the exam.

Please read the instructions carefully before proceeding.
